

AfriSalaries

ML-Powered Salary Intelligence for African Tech

Complete Project Documentation

Data Collection · Model Development · Architecture Decisions · Honest Evaluation

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Project Statement & Vision

African tech is one of the fastest-growing talent markets on the planet. It is also one of the most opaque when it comes to compensation. Over 70% of tech job posts across Africa include no salary information. That is not accidental. It is structural.

Companies maintain internal salary bands and negotiation playbooks. Candidates walk into every discussion blind — armed with whatever they heard from a colleague, if anything at all. The consequences are real.

A junior engineer in Nairobi accepts KES 80,000/month without knowing the market rate for her role is KES 140,000. An HR team in Amsterdam sets pay for a Lagos remote hire using European benchmarks and gets it wrong. A startup in Accra loses candidates to a competitor that was simply transparent about what the job pays. A Kenyan diaspora professional returning home cannot benchmark what to ask for after years earning in GBP.

AfriSalaries is a machine learning system that takes any tech job description as plain text and returns a predicted salary band — LOW, MEDIUM, or HIGH relative to the local market — with a full explainable breakdown of which factors drove the estimate. No account required. No form. Paste the description. Get the band. Understand why.

Scope

2.1 Geographic Scope

COUNTRY	CODE	PRIMARY HUB	CURRENCY
Kenya	KE	Nairobi	KES
Nigeria	NG	Lagos, Abuja	NGN
South Africa	ZA	Cape Town, Johannesburg	ZAR
Ghana	GH	Accra	GHS
Ethiopia	ET	Addis Ababa	ETB
Tanzania	TZ	Dar es Salaam	TZS
Uganda	UG	Kampala	UGX
Rwanda	RW	Kigali	RWF

2.2 Domain Scope

AfriSalaries covers technology and software roles only in v1.0. That includes software engineering (backend, frontend, full-stack, mobile), data science, machine learning, AI engineering, DevOps, cloud engineering, product management, UX/UI design, and solutions architecture. Non-tech roles, hardware, academic research positions, and government civil service are explicitly excluded. Their compensation structures are incomparable.

2.3 Technical Architecture

LAYER	TECHNOLOGY	PURPOSE
ML Model	XGBoost + TF-IDF + SHAP	Band classification + explainability
API	FastAPI + Pydantic v2	REST endpoint, auto /docs
Database	Neon PostgreSQL (async)	Prediction history
Cache	Upstash Redis REST	Repeated query optimisation
Frontend	React 18 + Vite + Tailwind	User dashboard
CI/CD	GitHub Actions	Automated testing on push
Backend Host	Render (free tier)	uvicorn, auto-redeploy
Frontend Host	Vercel (free tier)	CDN-backed, zero-config
Total Cost	\$0/month	Fully serverless free tier

Objectives

3.1 Primary Objectives

Objective 1 — Build a working salary intelligence API.

Develop, train, and deploy a machine learning model that accepts a tech job description as input and returns a salary band prediction with confidence score in under 200 milliseconds.

Objective 2 — Achieve explainability for every prediction.

Every estimate must be accompanied by a SHAP-based breakdown showing which words and features drove the classification. The system must be auditable and trustworthy, not a black box.

Objective 3 — Cover all 8 target countries.

The model must produce meaningful predictions for all 8 African countries, not only data-rich markets like South Africa and Nigeria.

3.2 Secondary Objectives

Objective 4 — Establish a real public salary dataset.

Publish the cleaned, anonymised dataset under CC BY-SA 4.0 as the first openly available tech salary dataset covering 8 African countries simultaneously.

Objective 5 — Create a feedback loop for continuous improvement.

The POST /predict/feedback endpoint allows users to submit salary corrections that become retraining data — making the model progressively more accurate.

Objective 6 — Demonstrate production ML from Africa.

Prove that production-grade, explainable, deployed ML systems can be built from Africa, on African data, for African users — without a GPU, without a lab, without Western

resources.

Honest Limitations

Intellectual honesty is a prerequisite for credibility. The following limitations are documented not as apologies but as scientific disclosures. A model that knows its limits is more trustworthy than one that pretends to know everything.

L1 — Stack Overflow User Bias

SO users skew toward experienced, English-speaking, internet-connected developers. Junior developers in peri-urban areas and developers without SO accounts are underrepresented. The model may overestimate salaries for entry-level roles in less-connected markets.

L2 — Sparse Country Coverage

Rwanda (28 rows), Tanzania (32), Uganda (48), Ethiopia (54), Ghana (60) all fall below the 100-row threshold for reliable country-specific learning. Predictions for these markets carry elevated uncertainty.

L3 — Self-Reported Salary Bias

All salary data is self-reported. Respondents who feel well-paid are more likely to share. The dataset likely has an upward salary bias of 5–15%.

L4 — Exchange Rate Sensitivity

The Nigerian naira devalued approximately 50% in 2023. Rows from 2022 converted at pre-devaluation rates may not reflect current USD purchasing power.

L5 — Base Salary Only

The model predicts base salary. Equity, bonuses, and benefits are not captured. A \$30,000 salary with 5% equity at a Nairobi Series B startup is economically different from \$30,000 at a government parastatal.

L6 — Cannot Be Used for Discrimination

The model must never be used to set different salary bands based on gender, nationality, ethnicity, or age. These variables are excluded by design, consistent with Kenya DPA 2019 and GDPR.

L7 — Not a Legal Instrument

AfriSalaries predictions are statistical estimates, not legal salary benchmarks. They cannot be cited in employment contracts or regulatory filings as authoritative data.

The Questions

Before writing a single line of model code, the following questions were posed and answered. This discipline separates rigorous ML practice from casual experimentation.

5.1 Questions About the Data

QUESTION	ANSWER
What is the unit of salary?	CompTotal captures total compensation. CompFreq specifies Yearly/Monthly/Weekly. All converted to annual USD — Monthly $\times 12$, Weekly $\times 52$.
Is salary gross or net?	Self-reported surveys almost always capture gross. Tax rates vary: South Africa up to 45%, Kenya up to 35%, Nigeria up to 24%. AfriSalaries reports gross and notes this clearly.
What is the target variable distribution?	Salary is log-normally distributed. Mean \$37,268 vs std \$33,797 confirms extreme skew. log1p transformation applied before training. Back-transform with expm1() for output.
Are there duplicates?	Rows where both description AND salary were identical were removed. Kept rows with same salary but different descriptions — legitimate market variation.

5.2 Questions About the Problem

QUESTION	ANSWER
Regression or classification?	Originally attempted regression. After two failed attempts with MAPE > 87%, pivoted to classification — predict salary band LOW/MEDIUM/HIGH. See Chapter 12.
What counts as a good prediction?	MAPE is the primary metric. A \$5,000 error on \$30,000 (17%) is worse than \$5,000 on \$150,000 (3%). Target: MAPE < 25% for regression, accuracy > 60% for classification.

Who is the primary user?	(1) African tech job seekers validating an offer. (2) International HR teams setting pay for African remote hires. (3) Researchers and policy makers accessing the public dataset.
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5.3 Questions About the Algorithm

QUESTION	ANSWER
Is the relationship linear?	No. XGBoost's tree-based structure learns non-linear interactions — country × seniority × skill — without explicit engineering of every combination. Linear models failed this requirement.
Do we need interpretability?	Yes. Non-negotiable. Without SHAP, the tool is a black box. With SHAP TreeExplainer, every prediction comes with ranked factors. This rules out neural networks.
Do we have enough data for deep learning?	No. Rule of thumb: XGBoost optimal at 1k–100k rows, neural networks minimum 10k–50k. Our 1,371 rows puts us squarely in the XGBoost range.
How often will the model retrain?	Quarterly retraining targeted for v1.x. Salary data drifts meaningfully — the 2022 NGN figures already reflect a different economic reality than 2025.

Data Sources & Licensing

6.1 Primary Source — Stack Overflow Developer Survey

The Stack Overflow Annual Developer Survey is the world's largest survey of software developers, conducted yearly since 2011. Stack Overflow publishes the full anonymised dataset under Creative Commons CC BY-SA 4.0, permitting unrestricted use including commercial applications, provided attribution is maintained.

ATTRIBUTE	DETAIL
Provider	Stack Overflow (Stack Exchange company)
License	Creative Commons CC BY-SA 4.0
URL	survey.stackoverflow.co
Years Used	2022, 2023, 2024, 2025
Global Respondents	~277,000 across all 4 years
Cost	Free. No registration required.
Contains PII	No. All data is anonymised.

6.2 Attempted Additional Sources — All Failed

Extensive efforts were made to supplement the Stack Overflow data with additional real sources. Every attempt is documented below with outcome and reasoning for exclusion.

SOURCE	ATTEMPT	OUTCOME	DECISION
ILO ILOSTAT Bulk	Direct .gz download	HTTP 404	URL changed — excluded

ILO API per country	REST API call	HTTP 400 all 5 countries	API format mismatch — excluded
World Bank LFS	Microdata portal	Pre-2017 data	Too old for 2026 model — excluded
GitHub community repos	5 repos attempted	HTTP 404 all	Repos deleted/renamed — excluded
Humanitarian Data API	HDX API	HTTP 404	Resource not found — excluded
AfricaOpenData	REST API	HTTP 403	Access restricted — excluded
Andela reports	PDF extraction	Not attempted	Remote-worker bias too severe — excluded by design

Conclusion: African salary data remains inaccessible through automated means as of May 2026. This is itself a research finding. The Stack Overflow survey, despite its representational limitations, is the only clean, licensed, programmatically accessible source of individual-level tech salary data for African countries currently available.

Acquisition Log — 4,911 Rows Across 4 Years

Four years of Stack Overflow survey data were downloaded programmatically from the GitHub StackExchange/Survey repository. URLs changed in early 2026 from Sanity CDN to direct GitHub hosting — a change discovered through live testing.

SURVEY YEAR	GLOBAL RESPONDENTS	AFRICAN ROWS
2025	49,191	734
2024	65,437	1,062
2023	89,184	1,563
2022	73,268	1,552
TOTAL	277,080	4,911

Country Distribution — Raw (Pre-Cleaning)

COUNTRY	COD E	2025	2024	2023	2022	TOTAL RAW
South Africa	ZA	253	358	573	517	1,701
Nigeria	NG	199	305	447	493	1,444
Kenya	KE	148	180	244	270	842
Ghana	GH	46	70	90	86	292
Ethiopia	ET	38	60	86	75	259
Uganda	UG	28	39	60	63	190
Tanzania	TZ	7	28	36	28	99

Rwanda	RW	15	22	27	20	84
TOTAL	—	734	1,062	1,563	1,552	4,911

Cleaning Pipeline — 4,911 to 1,371

Stage 1 — Null Salary Removal: 4,911 → ~2,800

The salary question (CompTotal) is entirely optional in the Stack Overflow survey. Approximately 43% of African respondents left it blank. Reasons include cultural norms around salary privacy, or the respondent is a student. Rows with null, zero, or non-numeric CompTotal were dropped.

Stage 2 — Currency Conversion Failures: ~2,800 → ~2,400

Every remaining salary was converted to annual USD using World Bank average exchange rates. Rates used: KES 130, NGN 1500 (post-devaluation), ZAR 18.5, GHS 12.5, ETB 57, TZS 2500, UGX 3700, RWF 1300. Rows were dropped when currency was null, blank, unrecognised, or when an African respondent reported in EUR/GBP.

Stage 3 — Sanity Bounds: ~2,400 → ~1,800

Hard bounds applied: minimum \$3,000 USD/year, maximum \$200,000 USD/year. Lower bound removes misreported hourly or daily rates entered as annual figures. Upper bound removes total compensation including vested equity and expatriate packages that would distort predictions for local market users.

Stage 4 — Deduplication: ~1,800 → 1,675

Rows where BOTH description AND salary_usd were identical were removed, keeping the first occurrence. This eliminates respondents who completed the survey in multiple years with identical answers. Rows with same salary but different descriptions were kept — legitimate market variation.

Stage 5 — Country Balancing: 1,675 → 1,371

South Africa represented 59% of cleaned data (991 rows) due to higher Stack Overflow engagement. Training on this imbalanced data produces a model calibrated to Cape Town rates that systematically mispredicts for Nairobi and Lagos. Decision: ZA capped at 3× Nigeria's count. Nigeria had 231 rows, so ZA capped at 693. Factor of 3 preserves ZA's highest-salary market position without drowning other markets.

Final Dataset Summary

COUNTRY	CODE	FINAL ROWS	QUALITY FLAG
South Africa	ZA	693	Adequate
Nigeria	NG	231	Adequate
Kenya	KE	225	Adequate
Ghana	GH	60	Adequate
Ethiopia	ET	54	High Uncertainty
Uganda	UG	48	High Uncertainty
Tanzania	TZ	32	High Uncertainty
Rwanda	RW	28	High Uncertainty
TOTAL	—	1,371	—

Salary statistics of final dataset: Min \$3,000 | Median \$27,243 | Mean \$37,149 | Max \$200,000 | Std Dev \$33,845.

Feature Engineering

9.1 Description Construction

Each row's description string is assembled from structured survey columns across all four survey years, which have different column names — handled explicitly. Components: seniority level, years of experience, developer role, programming languages (up to 4), cloud platforms (up to 2), databases (up to 2), remote/hybrid status, city and country name. Average description length: 129 characters.

Example output: "Senior 8+ years Developer, full-stack Go HTML/CSS JavaScript Kotlin Firebase Google Cloud Remote Cape Town South Africa"

9.2 Features Kept — With Justification

FEATURE	TYPE	WHY ESSENTIAL
Seniority level	Numerical 0.0–1.0	Single highest predictor — doubles salary from Junior to Senior
Years of experience	Numerical	Strong signal within seniority bands
Developer role	Text	ML Engineer vs IT Support is a 3× salary difference
Programming languages	Text	Python/Go/Rust command premium over PHP/VBA
Cloud platforms	Text	AWS/GCP/Azure each add measurable salary premium
Remote status	Binary flag	Remote adds 15–30% in African markets
Country	One-hot encoded	Nairobi vs Cape Town = 40% salary difference, same role

High-value skill count	Numerical	Count of 22 premium skills (Kubernetes, PyTorch, etc.)
Country baseline	Numerical	Known market-level salary anchor per country
Fintech signal	Binary flag	Fintech pays 30–50% above NGO for same role
Management signal	Binary flag	Director/VP/CTO commands significant premium

9.3 Features Dropped — With Justification

FEATURE	REASON DROPPED
Company name	Too sparse — most appear once. Causes overfitting.
Gender	Illegal — Kenya DPA 2019 and GDPR prohibit use in salary prediction.
Age	Illegal — age discrimination laws in all 8 countries.
Education level	Rarely in job posts. Too many nulls at inference time.
City within country	Too sparse. Country is sufficient. Nairobi vs Mombasa needs 100+ rows/city.
Exact posting date	Not available at inference time.
Survey year	Too few rows to learn year-on-year drift reliably.
Raw currency	Always convert to USD first. Currency is an artifact of geography.

Model v1 — Baseline Failure & Diagnosis

The first model was a standard XGBoost regression with TF-IDF text features and country one-hot encoding. It represented the most natural first approach to salary prediction: predict the exact salary value.

10.1 Configuration

TF-IDF: 5,000 features, bigrams, sublinear_tf=True, min_df=2. Country one-hot: 8 features. Combined: 5,008 total features. XGBoost: n_estimators=500, max_depth=6, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, min_child_weight=3, reg_alpha=0.1, reg_lambda=1.0.

10.2 Results

METRIC	v1 RESULT	TARGET	STATUS
R-squared (R2)	0.3388	>= 0.65	FAIL
MAE (USD)	\$19,514	< \$8,000	FAIL
MAPE	92.1%	< 25%	FAIL
Median AE	\$10,335	< \$5,000	FAIL
p50 Error	48.1%	< 20%	FAIL
p90 Error	181.6%	< 50%	FAIL

10.3 Critical Observation

Sample predictions revealed the model was predicting the mean. Nearly every prediction clustered between \$35,000–\$46,000 regardless of actual salary, which ranged from \$3,000 to \$200,000. This is the classic signature of a regression model that has learned insufficient

signal to discriminate — it defaults to the training mean as the safest prediction under high uncertainty.

Model v2 — Enhanced Features & Continued Failure

After diagnosing v1, three specific fixes were implemented: (1) `min_df` reduced from 2 to 1 to recover lost vocabulary, (2) eight handcrafted numerical features added, (3) hyperparameters adjusted for stronger regularisation.

11.1 New Features Added

FEATURE	TYPE	PURPOSE
seniority_score	Float 0.0–1.0	Converts seniority words to numbers
years_exp	Integer	Extracted from description via regex
is_remote	Binary	Remote flag for premium detection
hv_skill_count	Integer	Count of 22 high-value tech skills
country_baseline	Float (thousands USD)	Known market anchor per country
desc_length	Float	Proxy for job seniority complexity
is_fintech	Binary	Fintech industry premium signal
is_management	Binary	Management/director role signal

11.2 Results Comparison

METRIC	v1	v2	CHANGE	TARGET
R2	0.3388	0.3846	+0.0458	≥ 0.65
MAE	\$19,514	\$18,654	−\$860	< \$8,000

MAPE	92.1%	87.8%	−4.3pp	< 25%
TF-IDF Features	1,279*	2,057	+778	5,000

* v1 produced only 1,279 TF-IDF features despite requesting 5,000 because min_df=2 eliminated 75% of the vocabulary. This was a critical undetected bug in v1 configuration.

11.3 Conclusion

Despite meaningful improvements in all metrics, v2 remained far below the deployment threshold. The model still predicted in a narrow cluster around the mean. The marginal improvements from 8 additional features confirmed that the fundamental problem was not feature engineering — it was the prediction task itself.

Root Cause Analysis — Why Both Models Failed

Root Cause 1 — Salary Range Is Statistically Intractable

The training set spans \$3,000 to \$200,000 — a 66× range. A Senior Python developer in Kigali may earn \$18,000. An identical role in Cape Town may earn \$65,000. A remote Senior ML Engineer anywhere in Africa may earn \$120,000. The within-description variance exceeds the between-description variance. No text model with 1,371 rows can reliably bridge a 66× prediction range.

Root Cause 2 — Seniority Has No Variance

The `seniority_score` distribution showed `mean=0.74`, meaning the training dataset is dominated by Senior and Lead developers. Without variation in the most predictive feature, the model cannot learn the Junior-to-Senior salary gradient.

Root Cause 3 — Country Baseline Dominates Incorrectly

The `country_baseline` feature showed 50th and 75th percentile both equal to \$42,000 (South Africa's baseline). Despite balancing, ZA still anchors the mean. All predictions are pulled toward \$42,000 regardless of other features.

Root Cause 4 — 1,371 Rows Is Insufficient for 66× Variance

Statistical theory: to reliably predict a continuous target with coefficient of variation > 0.9 (our $CV = 0.91$), you need approximately 10,000+ training rows per major subgroup. We have 225 rows for Kenya and 28 for Rwanda. The model cannot learn country-specific salary gradients from these sample sizes.

The Strategic Decision — Why Salary Bands

Two options were considered after v2 failure.

Option A — Continue Fixing Data

Acquire more real data from ILO, World Bank, or Kaggle to increase sparse country representation and reduce the 66× salary range problem. This was attempted extensively over two days. Every external source returned HTTP 404, 400, or 403. African salary data remains inaccessible through automated means as of May 2026. Verdict: not viable within any reasonable timeframe.

Option B — Change the Task to Salary Band Prediction

Instead of predicting exact salary (regression), predict whether a salary is LOW, MEDIUM, or HIGH relative to the local market (classification). This reduces the 66× prediction range to three tractable buckets. Verdict: the correct statistical and business decision.

Why Band Prediction Is Not a Compromise

Salary bands are how every HR department in the world thinks. Glassdoor, Levels.fyi, and LinkedIn Salary all show ranges, not exact figures. A candidate needs to know 'Is this offer fair?' not 'The mathematically precise market rate is \$34,782.50.' Band prediction directly answers the real user question. An exact salary prediction with 87.8% MAPE is not a product — it is a liability. Telling someone earning \$30,000 that they should earn \$65,000 causes real harm in real salary negotiations.

Alignment With Personal Goals

GOAL

EXACT REGRESSION

BAND CLASSIFICATION

Top-earning employment	No deployed product = nothing to show	Live product + honest decisions = hireable
University admission	Perfect metrics look fabricated	Failed attempts + pivot = scientific maturity
SaaS viability	Harmful predictions = brand destruction	Honest bands = defensible product

The honest narrative of what was tried, why it failed, and how the decision to pivot was made is itself a research contribution. European universities do not want a student who claims perfect results. They want a student who demonstrates scientific rigour, intellectual honesty, and the maturity to change approach when evidence demands it.

Model v3 — Two-Stage Architecture

Version 3 implements a two-stage architecture that directly addresses the statistical intractability of single-model exact salary regression.

14.1 Architecture Overview

STAGE	MODEL	TASK	OUTPUT
Stage 1	XGBoost Classifier	Predict salary band	LOW / MEDIUM / HIGH
Stage 2	XGBoost Regressor (×3)	Predict salary within band	Salary range per band

14.2 Band Thresholds

BAND	SALARY RANGE (USD/YEAR)	MEANING FOR USER
LOW	< \$10,000	Below market — strong grounds to negotiate
MEDIUM	\$10,000 – \$35,000	At market — negotiate on specific factors
HIGH	> \$35,000	Above market — consider total compensation

14.3 Why This Works

Within each band, the salary range is compressed from 66× to approximately 3–5×. A LOW band regressor predicts between \$3,000 and \$10,000 — a 3.3× range that a text model with even 300 rows can learn meaningfully. The classifier only needs to answer one question: 'Is this description LOW, MEDIUM, or HIGH?' — a much simpler question than 'What is the exact annual salary of this person?'

Decision Log — All 7 Documented Decisions

Every significant decision in this project is recorded with date, decision statement, and full justification. This log serves as the project's scientific audit trail.

DECISION 1 · Use Only Real Data

May 19, 2026

No synthetic data will be used in the final training dataset under any circumstances. AfriSalaries makes claims about real African tech salary markets. Those claims must be grounded in real observations from real workers. A model trained on synthetic data cannot tell a Ugandan developer what the market pays — it can only reflect the assumptions of the data generator.

DECISION 2 · Three Sources Only: SO, World Bank, Kaggle

May 19, 2026

Andela data excluded: Andela-placed engineers earn 2–3× local rates due to international placement. Including this without clean separation would systematically overestimate salaries for Uganda, Rwanda, Ghana, and Ethiopia. World Bank preferred over ILO: World Bank provides individual-level microdata. ILO provides aggregated statistics — you cannot generate real rows from aggregates.

DECISION 3 · Minimum 30 Rows Per Source Per Country

May 19, 2026

No source included unless it contributes minimum 30 clean rows for at least one sparse country. Below 30 rows, the preprocessing overhead and licensing obligations exceed the benefit to model performance.

DECISION 4 · Source-Specific Salary Bounds

May 19, 2026

Stack Overflow: \$3,000–\$200,000. World Bank LFS: \$8,000–\$120,000 (excludes low-wage ICT clerks and expatriate packages). Kaggle: \$5,000–\$180,000. Different sources have different structural characteristics producing different outlier patterns.

DECISION 5 · Provenance Column Mandatory

May 19, 2026

Every row carries a source column (stackoverflow / worldbank / kaggle). Enables source-specific analysis, future exclusion if quality issues emerge, and honest licensing documentation.

DECISION 6 · World Bank ISCO Filter

May 19, 2026

Only ISCO-08 codes 2511–2529 (ICT professionals) and 1330 (ICT managers) included. ISCO 35xx (ICT technicians) excluded. AfriSalaries serves software developers and data professionals. Technicians earn \$5,000–\$12,000 and have fundamentally different career paths. Inclusion would drag predictions downward.

DECISION 7 · Pivot from Regression to Band Classification

May 20, 2026

After two regression model attempts (v1: $R^2=0.34$, MAPE=92.1%; v2: $R^2=0.38$, MAPE=87.8%), exact salary regression was declared statistically intractable with 1,371 rows spanning a \$3,000–\$200,000 range (CV=0.91). Root causes identified: 66× salary range, seniority variance collapse, country baseline dominance, insufficient rows per subgroup. Pivoted to two-stage architecture: band classifier + per-band regressor. This reflects industry standard practice and directly answers the user's negotiation question.

Next Steps & Roadmap

v1.0 — Complete

Data acquisition pipeline (SO survey 2022–2025). Cleaning pipeline (4,911 → 1,371 real rows). Feature engineering (TF-IDF + 8 numerical features). Two failed regression models, diagnosed. Two-stage band classification architecture (v3). All decisions documented with justification.

v1.x — Next Immediate Steps

Train v3 band classifier, record accuracy and CV scores. Commit model to repo (band_classifier.pkl, band_regressors.pkl). Deploy Render API at afrisalaries-api.onrender.com. Deploy Vercel frontend at afrisalaries.vercel.app. Set up Neon PostgreSQL and Upstash Redis credentials. Write honest README with v3 metrics and limitations.

v2.x — Medium Term

Real training data from BrighterMonday + Fuzu scraper (3,000+ rows). Return to salary regression once data exceeds 5,000 rows per country. Currency conversion with year-specific exchange rates. Browser extension to auto-predict on any African job board page. Weekly model retraining from accumulated feedback data.

v3.x — Long Term

Company-level salary models for the top 20 African tech employers. Peer-reviewed publication on African tech salary transparency. Expansion to 16 African countries. SMS salary benchmarking via Africa's Talking API.

Personal Goals & Portfolio Value

This chapter is written directly to the three audiences who will evaluate this work: technical employers, European university admissions committees, and SaaS users. The same project speaks differently to each.

17.1 For Technical Employers and Remote Clients

The question a hiring manager asks is not 'did the model work?' It is: 'Does this person understand what they are doing and why?' AfriSalaries answers that question clearly.

This project demonstrates end-to-end ML system design from data acquisition to deployment; honest scientific method — two failed attempts documented, diagnosed, and corrected; production engineering with FastAPI, Docker, CI/CD, Render, Vercel, PostgreSQL, Redis; data ethics embedded in every decision; and the rarest engineering quality of all — the willingness to say 'this approach failed' and explain precisely why, rather than shipping a broken model and hoping nobody notices.

17.2 For European University Admissions

AfriSalaries is, at its core, a research project in African labour market economics and applied natural language processing. The documentation you are reading — the questions asked, the hypotheses tested, the failures encountered, the pivot justified — is a research methodology section. It demonstrates scientific maturity that most applicants, including those from well-funded Western institutions, do not produce.

A student who says 'I trained a model and got $R^2=0.97$ ' raises immediate suspicion. A student who says 'I attempted exact salary regression, obtained $R^2=0.34$, diagnosed three structural root causes in the data, made a documented decision to pivot to ordinal classification, and deployed a working system with honest uncertainty quantification' — that student is ready for graduate-level research.

This project also represents a genuine research gap. A live, open, CC BY-SA licensed tech salary dataset covering 8 African countries does not currently exist in the public domain. AfriSalaries is creating it. That is an original contribution, not a class assignment.

17.3 For SaaS Viability

AfriSalaries' commercial proposition is simple and defensible: African tech workers lack the salary intelligence that their counterparts in London, Amsterdam, and New York take for granted. The market for this information is real and growing. The two-stage band prediction architecture produces results that are honest about their limitations and useful for their intended purpose — helping a candidate decide whether to negotiate, accept, or walk away from an offer.

Built with intellectual honesty in Kisumu, Kenya.
For the millions of African tech workers who deserve to know what their skills are worth.